

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

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Lab Assignment 5: Linear Search

Experiment No. 5.1 - Linear search for a key that is present

Experiment Title:

Linear search for a key that is present

Aim:

To search for the key 80 in an array using linear search.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Linear search checks the elements one by one from the first to the last. If the key matches an element, its index is returned; if the end is reached without a match, -1 is returned. It works on unsorted arrays and takes O(n) time in the worst case.

Algorithm / Procedure:

1. Write linearSearch(a, key) that loops over every index.
2. If a[i] == key, return i.
3. After the loop, return -1.
4. Call it with key 80 and print the result.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 5, Task 1
// Question: Write the method int linearSearch(int[] a, int key) that
// returns the index of key, or -1 when key is not in the array. Call
// it with {45, 67, 80, 52, 90} and key 80, and print the result.

public class Main {
    static int linearSearch(int[] a, int key) {
        for (int i = 0; i < a.length; i++) {
            if (a[i] == key) {
                return i;
            }
        }
        return -1;
    }

    public static void main(String[] args) {
        int[] a = {45, 67, 80, 52, 90};
        System.out.println(linearSearch(a, 80));
```

```
    }
}
```

Input:

Array {45, 67, 80, 52, 90}, key = 80.

Output:

```
2
```

Time / Space Complexity: Time O(n), best case O(1); Space O(1).

Result:

The program was compiled and executed successfully. The key 80 was found at index 2.

Viva-Voce Questions:

- What does linearSearch return when the key is missing?
- Does linear search need a sorted array?
- What is the best case of linear search?

Experiment No. 5.2 - Linear search for a key that is absent

Experiment Title:

Linear search for a key that is absent

Aim:

To show that linear search returns -1 when the key is not in the array.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Linear search checks the elements one by one from the first to the last. If the key matches an element, its index is returned; if the end is reached without a match, -1 is returned. It works on unsorted arrays and takes O(n) time in the worst case.

Algorithm / Procedure:

1. Write linearSearch(a, key) that loops over every index.
2. If a[i] == key, return i.
3. After the loop ends without a match, return -1.
4. Call it with key 100 and print the result.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 5, Task 2
// Question: Write the method int linearSearch(int[] a, int key) that
// returns the index of key, or -1 when key is not in the array. Call
// it with {45, 67, 80, 52, 90} and key 100, and print the result.

public class Main {
    static int linearSearch(int[] a, int key) {
        for (int i = 0; i < a.length; i++) {
            if (a[i] == key) {
                return i;
            }
        }
        return -1;
    }

    public static void main(String[] args) {
        int[] a = {45, 67, 80, 52, 90};
        System.out.println(linearSearch(a, 100));
    }
}
```

Input:

Array {45, 67, 80, 52, 90}, key = 100.

Output:

```
-1
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The key 100 is not in the array, so the method returned -1.

Viva-Voce Questions:

- Why is -1 used for 'not found'?
- How many comparisons are made for an absent key?
- What is the worst case of linear search?

Experiment No. 5.3 - Count occurrences of a value

Experiment Title:

Count occurrences of a value

Aim:

To count how many times the value 52 appears in an array.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Linear search checks the elements one by one from the first to the last. If the key matches an element, its index is returned; if the end is reached without a match, -1 is returned. It works on unsorted arrays and takes O(n) time in the worst case.

Algorithm / Procedure:

1. Create the array {52, 10, 52, 7, 52} and set count = 0.
2. Loop over every element.
3. If a[i] == 52, increase count.
4. Print count with the label Count.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 5, Task 3
// Question: Count how many times the value 52 appears in the array
// {52, 10, 52, 7, 52}. Print the count as "Count:" followed by the
// number.

public class Main {
    public static void main(String[] args) {
        int[] a = {52, 10, 52, 7, 52};
        int count = 0;

        for (int i = 0; i < a.length; i++) {
            if (a[i] == 52) {
                count++;
            }
        }

        System.out.println("Count: " + count);
    }
}
```

Input:

Array {52, 10, 52, 7, 52}, value = 52.

Output:

```
Count: 3
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The value 52 appears 3 times in the array.

Viva-Voce Questions:

- How is counting different from searching for the first match?
- Why must the loop visit every element?
- What is the time complexity?